

Education

Bachelor of Software Engineering (B. Seng) | University of Victoria | 2nd Year

Sept 2024 – Dec 2028

Relevant courses – Data Structures & Algorithms, Software Development Methods, Object Oriented Programming, Fundamentals of Programming (C, Java & Python).

GPA: 4.0 / 4.0

Technical Skills

Programming Languages: Python, C, Java, JavaScript, bash, SQL (MySQL), MATLAB, Thonny, R

Web development: HTML, CSS, Embedded JavaScript (EJS), RESTful APIs.

Frameworks & Libraries: Flask, FastAPI, Node.JS, React.js, Express.js

Databases & Data Tools: MySQL, PostgreSQL, SQLite, Microsoft Excel (VLOOKUP, PivotTables), Power BI, Microsoft Automate

Tools & Platforms: Git, GitHub, GitHub Actions (CI/CD), Docker, Playwright, VS Code, Jupyter Notebook, Markdown

Testing & QA: Unit Testing, Pytest, Edge-Case Coverage, Debugging, Data Validation, Fault-Tolerant Design

Embedded Systems & IoT: Raspberry Pi Pico, MicroPython, GPIO, Real-Time Data Acquisition.

Other: Agile Methodologies, Technical Documentation, Data Visualization

Technical Projects

Backend Developer | University of Victoria Satellite Design Club, UVIC

Jan 2025 - Present

- Developed a **Python-based scalable backend** using traceable, **fault-tolerant architecture** to **automate** handling of binary image / telemetry data from satellite systems, reducing data loss by 40%.
- Designed an **interrupt-driven system** with **semaphores** and concurrency principles to enable real-time processing, improving throughput by 60%.
- Automated **data reassembly** and **segmentation** via **file I/O** and **control structures** to ensure consistent **identifier tagging** across sessions.
- Developed a **pytest test suite** with edge-case coverage to increase backend stability, cutting regressions by 50%.
- Documented test protocols and quality checks using clear **technical writing** to streamline onboarding and cross-team **maintenance**.
- Collaborated with hardware / logic teams via **GitHub** and **agile practices** to resolve conflicts and accelerate feature deployment.
- Identified and **debugged edge cases** in live **data capture** with adaptive **problem-solving** to ensure 99% uptime during critical missions.

AI Job Applier (Resume & Cover Letter Automation) | Personal Project.

Sep 2025 – Present

- Building an **AI-driven job application automation tool** using **FastAPI**, **React**, and **PostgreSQL** to auto-fill applications, generate tailored resumes/cover letters, and track recruiter replies streamlining the application process.
- Integrating **OpenAI API** for personalized document generation and **Gmail API** for recruiter response tracking, increasing personalization and enabling real-time monitoring of application outcomes.
- Developing **analytics dashboard** displaying conversion rates, response rates, and interview pipeline, providing actionable insights to optimize job search strategies.
- Containerizing the application with **Docker** and **Docker Compose**, ensuring scalability, portability, and smooth deployment across environments.

DevSwipe – GitHub Discovery Platform | Personal Project

Aug 2025 - Present

- Building a Tinder-style **web app** with **Flask**, **SQLAlchemy**, and **GitHub API** (CI/CD pipelines) to simplify repository discovery for 500+ users.
- Implementing authentication via **Werkzeug hashing** and sessions to maintain 100% **data integrity** and **security**.
- Designing an **SQLite schema** with **1-to-many relationships** to enable personalized **user tracking** and swipes.
- Enhancing engagement with **swipe mechanics** and filters to boost average session time by 30%.

- Developed a **Flask-based full-stack** system with **Raspberry Pi Pico** and **MicroPython** to detect falls in real time, reducing alert delays by 40%.
- Integrated live sensor data via **GPIO** and **ADC modules** to trigger instant emergency notifications, improving response time by 50%.
- Designed a dynamic dashboard using **JavaScript, HTML & CSS** to visualize motion trends, enhancing caregiver monitoring efficiency.
- Tested **system reliability** through simulated fall scenarios to minimize false alarms, achieving 90% detection accuracy.
- Documented hardware-software protocols with clear **technical writing** to simplify future upgrades by engineering teams.

Hackathons

TrafficTongue-AI-Powered Honking Detection & Noise Reduction System | Hackathon Finalist

June 2024

- Programmed custom audio packet logic in **Java** to interface with the backend, ensuring efficient stream processing and real-time data handling.
- Developed **embedded C** firmware for USB audio streaming to enable real-time honking detection.
- Trained a **TensorFlow** model on **Python** audio pipelines to classify honk types with 85% accuracy.
- Built a **Node.js** dashboard with real-time visualizations to display honking heatmaps for city planners.

Professional Experience

Voting Information Officer | UVSS, University of Victoria, Canada.

March 2025

- Maintained **100%** accurate voter records for **5,000+ students** using **Excel** and UVSS policies.
- Resolved eligibility issues and supported **smooth participation** during peak election periods.
- Increased voter turnout by **15%** through in-person and digital outreach campaigns.
- Collaborated with staff under tight deadlines to ensure **transparency** and **integrity**.

Certifications

- **Introduction to Cybersecurity** — Cisco Networking Academy, (Planned Sept 2025)
- **CompTIA Cybersecurity Fundamentals** — CompTIA, (Planned Sept 2025)
- **Prompt Engineering for AI** — Udemy, Ongoing
- **Responsive Web Design (HTML, CSS)** — Udemy, December 2024
- **Python for Everybody** — University of Michigan (Coursera), May 2024
- **CS50: Introduction to Computer Science** — Harvard, June 2023
- **Guitar Grade Exam** — Trinity College London, 2018 & 2019

Scholarships & Awards

- **Certificate of Achievement** – UVic Math & Stats Dept. (2025)
- **Entrance Scholarship for Academic Excellence** – UVic (2024)
- **PRAYAS Youth Impact Award** – NGO PRAYAS (2023)

Extracurricular Activities

- Volunteer Educator with **PRAYAS NGO** since 2019, **teaching** underprivileged children and reinforcing the importance of **accessibility in learning**.
- Write and perform original poetry, building **creative expression**, **public speaking**, and **emotional intelligence**.
- Play drums and guitar as a self-taught musician, developing **discipline**, **coordination**, and creative **collaboration**.
- Edit personal videos and event montages, showcasing storytelling, **attention to detail**, and digital content skills.
- Compete in cricket, squash, and swimming, strengthening **teamwork**, **focus**, and **resilience** through sports.
- Learning **French** to enhance **bilingual communication** skills and align with Canada's official language **culture**.